BASIC SKILLS BOOKLET

Algebra Basic Skills

Learn it. Practise it. Show you can do it.

Name: _____ Class: _____

How to use this booklet

1 **Learn**
Read the notes and the worked examples. Each example shows every step.

2 **Practise**
Do **Practice A**, then **Practice B**. Check your answers at the back of the booklet and fix any mistakes.

3 **Skill Check**
When you're ready, do the Skill Check **on your own, without notes**. Your teacher will mark it.

My progress

	Skill	Practice A done	Practice B done	Skill Check (teacher)
1	**Collecting Like Terms** I can add and subtract like terms, e.g. $3a + 4a = 7a$	☐	☐	/ 8
2	**Multiplying Algebra** I can multiply terms, e.g. $2a \times 3b = 6ab$ and $x \times x = x^2$	☐	☐	/ 8
3	**Dividing Algebra** I can divide terms, e.g. $8x \div 2 = 4x$ and $6ab \div 2a = 3b$	☐	☐	/ 8
4	**Expanding Brackets** I can expand brackets, e.g. $3(x + 2) = 3x + 6$	☐	☐	/ 8
5	**Factorising into Brackets** I can factorise, e.g. $2x + 6 = 2(x + 3)$	☐	☐	/ 8
	Mixed Review (extra practice)	☐		Total / 40

Tip: the answers to all the **practice** questions are at the back. The Skill Checks are not — that's where you show what you can do!

1 Collecting Like Terms

LEARN

🎯 I can add and subtract like terms, e.g. $3a + 4a = 7a$

Like terms have **exactly the same letter part**. You can only add or subtract like terms.

Like terms ✔	NOT like terms ✘
$3a$ and $5a$	$3a$ and $5b$ (different letters)
x and $7x$	x and 7 (one has no letter)
$2x^2$ and x^2	x^2 and x (x^2 is not the same as x)
$4ab$ and ab	ab and a

Think of the letters as objects. 3 apples + 4 apples = 7 apples, so $3a + 4a = 7a$.
But 3 apples + 2 bananas is just 3 apples + 2 bananas, so $3a + 2b$ stays as $3a + 2b$.

A letter on its own means 1 of it. a is the same as $1a$. So $a + a + a = 3a$.

Worked examples

$3a + 4a$

3 lots of a plus 4 lots of a

$= 7a$

$6x - 2x$

6 take away 2 is 4

$= 4x$

$5y - y$

y means $1y$, and $5 - 1 = 4$

$= 4y$

$2a + 3b + 4a + b$

The a's: $2a + 4a = 6a$
The b's: $3b + b = 4b$

$= 6a + 4b$

$x^2 + 3x + 2x^2$

The x^2's: $x^2 + 2x^2 = 3x^2$
The x's: just $3x$

$= 3x^2 + 3x$

$7m + 2 - 3m$

The m's: $7m - 3m = 4m$
The numbers: just 2

$= 4m + 2$

Remember: the sign (+ or −) in **front** of a term belongs to that term. Highlight or circle each type of term in a different colour before you add them up.

1 Collecting Like Terms

PRACTICE A

Simplify by collecting like terms. Show your working in the space.

1 $a + a$

= _____

2 $a + a + a$

= _____

3 $2a + 3a$

= _____

4 $4x + 5x$

= _____

5 $3m + m$

= _____

6 $7b - 3b$

= _____

7 $6y - y$

= _____

8 $10x - 4x$

= _____

9 $2p + 3p + p$

= _____

10 $8c - 2c$

= _____

11 $x^2 + x^2$

= _____

12 $3x^2 + 4x^2$

= _____

✔ Check your answers at the back. Fix any you got wrong before moving on.

1 Collecting Like Terms

PRACTICE B

Simplify by collecting like terms. Some have two types of term. Show your working in the space.

1 $2a + 3b + 4a$

=

2 $5x + 2y + x$

=

3 $3a + 2b + a + 4b$

=

4 $4x + 3 + 2x$

=

5 $7m - 2m + 3n$

=

6 $5a + 2b - 3a$

=

7 $6x + 4 - 2x + 1$

=

8 $8x - 3x + 2y$

=

9 $4ab + 2ab$

=

10 $x^2 + 2x + 3x$

=

11 $2x^2 + x + x^2$

=

12 $3y^2 + 2y + y^2 + 4y$

=

✔ Check your answers at the back. Fix any you got wrong before moving on.

1 Collecting Like Terms

SKILL CHECK

Name: _____

Teacher mark: _____ / 8

Simplify by collecting like terms. Do this on your own — no notes. Show your working. Write your final answer on the line.

1 $4a + 3a$

= _____

2 $9x - 5x$

= _____

3 $b + b + b$

= _____

4 $3a + 2b + 5a$

= _____

5 $6x + 4y - 2x$

= _____

6 $2x^2 + 5x^2$

= _____

7 $x^2 + 4x + 3x$

= _____

8 $3x^2 + 2x + x^2 + 5x$

= _____

2 Multiplying Algebra

LEARN

🎯 I can multiply terms, e.g. $2a \times 3b = 6ab$ and $x \times x = x^2$

In algebra we **don't write the × sign**. Letters and numbers that are multiplied are just written next to each other, **number first**.

We write...	...it means
$3a$	$3 \times a$
ab	$a \times b$
$5xy$	$5 \times x \times y$
x^2	$x \times x$ (say "x squared")

How to multiply terms

1. **Multiply the numbers** together.
2. **Write the letters** after the number (in alphabetical order).
3. If the **same letter** appears twice, write it as a **square**: $x \times x = x^2$.

Worked examples

$4 \times a$

Number first, then the letter

$= 4a$

$a \times b$

Write the letters together

$= ab$

$2a \times 4$

Numbers: $2 \times 4 = 8$
Letters: a

$= 8a$

$2a \times 3b$

Numbers: $2 \times 3 = 6$
Letters: a and b

$= 6ab$

$x \times x$

The same letter twice is "squared"

$= x^2$

$3x \times 2x$

Numbers: $3 \times 2 = 6$
Letters: $x \times x = x^2$

$= 6x^2$

Remember: when you **multiply**, the numbers multiply ($3 \times 2 = 6$) — don't add them! And $x \times x$ is x^2, not $2x$.
Going further: $x \times x \times x = x^3$ ("x cubed"), and $x^2 \times x = x^3$.

2 Multiplying Algebra

PRACTICE A

Simplify. Show your working in the space.

1 $4 \times a$

=

2 $5 \times x$

=

3 $a \times b$

=

4 $x \times y$

=

5 $2 \times 3a$

=

6 $3x \times 4$

=

7 $3 \times 5m$

=

8 $6 \times 2x$

=

9 $a \times a$

=

10 $y \times y$

=

11 $2a \times b$

=

12 $p \times q \times r$

=

✔ Check your answers at the back. Fix any you got wrong before moving on.

2 Multiplying Algebra

PRACTICE B

Simplify. Watch for the same letter twice! Show your working in the space.

1 $2a \times 3b$

=

2 $4x \times 2y$

=

3 $5a \times 2b$

=

4 $2ab \times 3$

=

5 $3a \times a$

=

6 $2x \times 5x$

=

7 $4m \times 3m$

=

8 $3x \times 2x$

=

9 $a \times 2b \times 3$

=

10 $x \times x \times x$

=

11 $x^2 \times x$

=

12 $2x^2 \times 3x$

=

✔ Check your answers at the back. Fix any you got wrong before moving on.

2 Multiplying Algebra

SKILL CHECK

Name: _____

Teacher mark: _____ / 8

Simplify. Do this on your own — no notes. Show your working. Write your final answer on the line.

1 $3 \times 4a$ ☐

= _____

2 $m \times n$ ☐

= _____

3 $4x \times 3$ ☐

= _____

4 $a \times 3b$ ☐

= _____

5 $2a \times 5b$ ☐

= _____

6 $x \times x$ ☐

= _____

7 $3y \times 2y$ ☐

= _____

8 $2x \times 4x$ ☐

= _____

3 Dividing Algebra

LEARN

🎯 I can divide terms, e.g. $8x \div 2 = 4x$ and $6ab \div 2a = 3b$

Dividing is the opposite of multiplying. A division can also be written as a fraction: $6a \div 2$ is the same as $\frac{6a}{2}$.

How to divide terms

1. **Divide the numbers.**
2. **Cancel any letter** that is on the top **and** the bottom.
3. Write down what is left.

Why can letters cancel? Anything divided by itself is 1. Just like $5 \div 5 = 1$, $a \div a = 1$. So $\frac{ab}{a} = b$.

Worked examples

$6a \div 2$

Numbers: 6 ÷ 2 = 3
Letters: a stays
$= 3a$

$ab \div a$

The a cancels
b is left
$= b$

$8x \div x$

The x cancels
8 is left
$= 8$

$12ab \div 4a$

Numbers: 12 ÷ 4 = 3
a cancels, b is left
$= 3b$

$x^2 \div x$

x^2 is $x \times x$
One x cancels, one is left
$= x$

$6x^2 \div 3x$

Numbers: 6 ÷ 3 = 2
Letters: $x^2 \div x = x$
$= 2x$

Remember: if **all** the letters cancel you are left with just a number, e.g. $5a \div a = 5$. Write x^2 as $x \times x$ if it helps you see what cancels.

3 Dividing Algebra

PRACTICE A

Simplify. Show your working in the space.

1 $8a \div 2$

=

2 $12x \div 3$

=

3 $10b \div 5$

=

4 $20y \div 4$

=

5 $9x \div 3$

=

6 $15p \div 5$

=

7 $18x \div 9$

=

8 $6m \div 6$

=

9 $ab \div b$

=

10 $xy \div x$

=

11 $5a \div a$

=

12 $7x \div x$

=

✔ Check your answers at the back. Fix any you got wrong before moving on.

3 Dividing Algebra

PRACTICE B

Simplify. Cancel the letters that are on top and bottom. Show your working in the space.

1 $6ab \div 2a$

=

2 $12xy \div 4y$

=

3 $10ab \div 5b$

=

4 $8xy \div 2x$

=

5 $14ab \div 7$

=

6 $x^2 \div x$

=

7 $a^2 \div a$

=

8 $4x^2 \div x$

=

9 $6x^2 \div 2x$

=

10 $15a^2 \div 5a$

=

11 $20x^2 \div 4x$

=

12 $9x^2 \div 3x$

=

✔ Check your answers at the back. Fix any you got wrong before moving on.

3 Dividing Algebra

SKILL CHECK

Name: _____

Teacher mark: _____ / 8

Simplify. Do this on your own — no notes. Show your working. Write your final answer on the line.

1 $10a \div 2$ ☐

= _____

2 $16x \div 4$ ☐

= _____

3 $xy \div y$ ☐

= _____

4 $9a \div a$ ☐

= _____

5 $8ab \div 4a$ ☐

= _____

6 $y^2 \div y$ ☐

= _____

7 $12x^2 \div 3x$ ☐

= _____

8 $20ab \div 5b$ ☐

= _____

4 Expanding Brackets

LEARN

🎯 I can expand brackets, e.g. $3(x + 2) = 3x + 6$

Expanding means getting rid of the brackets by multiplying. A number or letter right in front of a bracket means **multiply everything inside** the bracket by it.

Part 1: One bracket

Multiply the outside by EVERY term inside. Draw arrows so you don't miss one!

$3(x + 2) \rightarrow 3 \times x$ and $3 \times 2 \rightarrow 3x + 6$

$4(a + 5)$

$4 \times a = 4a$

$4 \times 5 = 20$

$= 4a + 20$

$2(x - 3)$

$2 \times x = 2x$

$2 \times 3 = 6$, keep the minus

$= 2x - 6$

$x(x + 5)$

$x \times x = x^2$

$x \times 5 = 5x$

$= x^2 + 5x$

Part 2: Two brackets

Everything in the first bracket multiplies everything in the second bracket. A **grid** makes it easy to keep track. Example: $(x + 2)(x + 3)$

×	x	+3
x	x^2	$3x$
+2	$2x$	6

1. Put one bracket on top and one down the side.
2. Multiply to fill each box.
3. Add up all four boxes: $x^2 + 3x + 2x + 6$
4. Collect the like terms in the middle:

$(x + 2)(x + 3) = x^2 + 5x + 6$

Remember: the outside term multiplies **every** term inside — $3(x + 2)$ is $3x + 6$, **not** $3x + 2$. With two brackets, you should always get **4** boxes.

4 Expanding Brackets

PRACTICE A

Expand. Show your working in the space.

1 $2(x + 3)$

= _____

2 $3(a + 4)$

= _____

3 $5(x + 1)$

= _____

4 $6(x + 2)$

= _____

5 $4(y - 2)$

= _____

6 $2(m - 5)$

= _____

7 $3(2x + 1)$

= _____

8 $2(4a + 3)$

= _____

9 $5(2y - 1)$

= _____

10 $x(x + 2)$

= _____

11 $a(a + 5)$

= _____

12 $x(x - 3)$

= _____

✔ Check your answers at the back. Fix any you got wrong before moving on.

4 Expanding Brackets

PRACTICE B

Expand and simplify. Use the grid to help. Show your working in the space.

1 $(x + 1)(x + 2)$

=

2 $(x + 3)(x + 4)$

=

3 $(x + 2)(x + 5)$

=

4 $(x + 1)(x + 6)$

=

5 $(x + 2)(x + 4)$

=

6 $(x + 5)(x + 1)$

=

7 $(x + 3)(x + 3)$

=

8 $(x + 4)(x + 5)$

=

✔ Check your answers at the back. Fix any you got wrong before moving on.

4 Expanding Brackets

SKILL CHECK

Name: _____

Teacher mark: _____ / 8

Expand (and simplify where needed). Do this on your own — no notes. Show your working. Write your final answer on the line.

1 $4(x + 2)$ ☐

= _____

2 $3(a - 5)$ ☐

= _____

3 $2(3x + 1)$ ☐

= _____

4 $5(2a + 3)$ ☐

= _____

5 $x(x + 7)$ ☐

= _____

6 $y(y - 4)$ ☐

= _____

7 $(x + 2)(x + 3)$ ☐

= _____

8 $(x + 1)(x + 4)$ ☐

= _____

5 Factorising into Brackets

LEARN

🎯 I can factorise, e.g. $2x + 6 = 2(x + 3)$

Factorising is the **opposite of expanding**. We put the brackets back in.

$2(x + 3)$ — expand → $2x + 6$ $\quad$ $2x + 6$ — factorise → $2(x + 3)$

Part 1: Take out a common factor

1. Find the **biggest number** (and/or **letter**) that goes into **both** terms. Write it outside the bracket.
2. **Divide** each term by it. Write the answers inside the bracket.
3. **Check** by expanding — you should get back to the question.

$5a + 10$

5 goes into both
$5a \div 5 = a$, $10 \div 5 = 2$

$= 5(a + 2)$

$6x - 9$

3 goes into both
$6x \div 3 = 2x$, $9 \div 3 = 3$

$= 3(2x - 3)$

$x^2 + 4x$

x goes into both
$x^2 \div x = x$, $4x \div x = 4$

$= x(x + 4)$

Part 2: Two brackets

To factorise $x^2 + 5x + 6$, find **two numbers** that **multiply** to the last number (6) and **add** to the middle number (5).

Multiply to 6	Add to…
1 × 6	1 + 6 = 7 ✘
2 × 3	**2 + 3 = 5 ✔**

The two numbers are 2 and 3, so they go in the brackets:

$x^2 + 5x + 6 = (x + 2)(x + 3)$

Check by expanding:
$(x + 2)(x + 3) = x^2 + 3x + 2x + 6 = x^2 + 5x + 6$ ✔

Remember: always **check by expanding** your answer. The brackets can go in either order: $(x + 3)(x + 2)$ is also correct.

5 Factorising into Brackets

PRACTICE A

Factorise. Show your working in the space.

1 $3x + 6$

=

2 $2a + 8$

=

3 $5x + 15$

=

4 $7m + 7$

=

5 $4y - 12$

=

6 $2x - 10$

=

7 $6a + 9$

=

8 $10x + 15$

=

9 $x^2 + 3x$

=

10 $a^2 + 5a$

=

11 $x^2 - 2x$

=

12 $y^2 + y$

=

✔ Check your answers at the back. Fix any you got wrong before moving on.

5 Factorising into Brackets

PRACTICE B

Factorise into two brackets. Show your working in the space.

1 $x^2 + 3x + 2$

= _____

2 $x^2 + 4x + 3$

= _____

3 $x^2 + 6x + 8$

= _____

4 $x^2 + 5x + 4$

= _____

5 $x^2 + 7x + 10$

= _____

6 $x^2 + 7x + 12$

= _____

7 $x^2 + 8x + 15$

= _____

8 $x^2 + 6x + 9$

= _____

✔ Check your answers at the back. Fix any you got wrong before moving on.

5 Factorising into Brackets

SKILL CHECK

Name: _____

Teacher mark: _____ / 8

Factorise fully. Do this on your own — no notes. Show your working. Write your final answer on the line.

1 $4x + 8$ ☐

= _____

2 $5m + 5$ ☐

= _____

3 $3a - 9$ ☐

= _____

4 $6y + 10$ ☐

= _____

5 $x^2 + 6x$ ☐

= _____

6 $a^2 - 4a$ ☐

= _____

7 $x^2 + 5x + 6$ ☐

= _____

8 $x^2 + 6x + 5$ ☐

= _____

Mixed Review

EXTRA PRACTICE

A mix of all five skills. Look at each question carefully — which skill do you need? Answers are at the back.

1. $5a + 2a$ = _____

2. $3 \times 2x$ = _____

3. $12y \div 3$ = _____

4. $2(x + 5)$ = _____

5. $3x + 12$ = _____

6. $4x + 3y + 2x$ = _____

7. $a \times 4b$ = _____

8. $10ab \div 2b$ = _____

9. $x(x + 1)$ = _____

10. $x^2 + 7x$ = _____

11. $8m - 3m$ = _____

12. $5x \times x$ = _____

13. $6x^2 \div x$ = _____

14. $4(2a - 1)$ = _____

15. $5y - 20$ = _____

16. $2x^2 + 3x + x^2$ = _____

17. $4a \times 2a$ = _____

18. $16xy \div 8x$ = _____

19. $(x + 1)(x + 3)$ = _____

20. $x^2 + 4x + 4$ = _____

Answers to the Practice Questions

Terms can be in a different order and still be right — for example $2b + 3a$ is the same as $3a + 2b$, and $(x + 3)(x + 2)$ is the same as $(x + 2)(x + 3)$.

1. Collecting Like Terms

Practice A

1. $2a$
2. $3a$
3. $5a$
4. $9x$
5. $4m$
6. $4b$
7. $5y$
8. $6x$
9. $6p$
10. $6c$
11. $2x^2$
12. $7x^2$

Practice B

1. $6a + 3b$
2. $6x + 2y$
3. $4a + 6b$
4. $6x + 3$
5. $5m + 3n$
6. $2a + 2b$
7. $4x + 5$
8. $5x + 2y$
9. $6ab$
10. $x^2 + 5x$
11. $3x^2 + x$
12. $4y^2 + 6y$

2. Multiplying Algebra

Practice A

1. $4a$
2. $5x$
3. ab
4. xy
5. $6a$
6. $12x$
7. $15m$
8. $12x$
9. a^2
10. y^2
11. $2ab$
12. pqr

Practice B

1. $6ab$
2. $8xy$
3. $10ab$
4. $6ab$
5. $3a^2$
6. $10x^2$
7. $12m^2$
8. $6x^2$
9. $6ab$
10. x^3
11. x^3
12. $6x^3$

3. Dividing Algebra

Practice A

1. $4a$
2. $4x$
3. $2b$
4. $5y$
5. $3x$
6. $3p$
7. $2x$
8. m
9. a
10. y
11. 5
12. 7

Practice B

1. $3b$
2. $3x$
3. $2a$
4. $4y$
5. $2ab$
6. x
7. a
8. $4x$
9. $3x$
10. $3a$
11. $5x$
12. $3x$

4. Expanding Brackets

Practice A

1. $2x + 6$
2. $3a + 12$
3. $5x + 5$
4. $6x + 12$
5. $4y - 8$
6. $2m - 10$
7. $6x + 3$
8. $8a + 6$
9. $10y - 5$
10. $x^2 + 2x$
11. $a^2 + 5a$
12. $x^2 - 3x$

Practice B

1. $x^2 + 3x + 2$
2. $x^2 + 7x + 12$
3. $x^2 + 7x + 10$
4. $x^2 + 7x + 6$
5. $x^2 + 6x + 8$
6. $x^2 + 6x + 5$
7. $x^2 + 6x + 9$
8. $x^2 + 9x + 20$

5. Factorising into Brackets

Practice A

1. $3(x + 2)$
2. $2(a + 4)$
3. $5(x + 3)$
4. $7(m + 1)$
5. $4(y - 3)$
6. $2(x - 5)$
7. $3(2a + 3)$
8. $5(2x + 3)$
9. $x(x + 3)$
10. $a(a + 5)$
11. $x(x - 2)$
12. $y(y + 1)$

Practice B

1. $(x + 1)(x + 2)$
2. $(x + 1)(x + 3)$
3. $(x + 2)(x + 4)$
4. $(x + 1)(x + 4)$
5. $(x + 2)(x + 5)$
6. $(x + 3)(x + 4)$
7. $(x + 3)(x + 5)$
8. $(x + 3)(x + 3)$

★ Mixed Review

1. $7a$
2. $6x$
3. $4y$
4. $2x + 10$
5. $3(x + 4)$
6. $6x + 3y$
7. $4ab$
8. $5a$
9. $x^2 + x$
10. $x(x + 7)$
11. $5m$
12. $5x^2$
13. $6x$
14. $8a - 4$
15. $5(y - 4)$
16. $3x^2 + 3x$
17. $8a^2$
18. $2y$
19. $x^2 + 4x + 3$
20. $(x + 2)(x + 2)$